SIMATS ENGINEERING
ASSESSMENT TOOL 2
Case study

COURSE NAME: Computer Networks **COURSE CODE: CSA07**

COURSE OUTCOME COVERED

CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*

Assessment Tool Weightage: 40%

Case study

QUESTIONS:4 **Total Marks:40**

1.A multinational IT company uses a cloud-based video conferencing platform for daily client meetings. During peak business hours, employees experience frequent voice distortion, frozen video frames, and delayed communication. Network monitoring reports indicate an **8% packet loss**, **180 ms jitter**, and fluctuating bandwidth utilization. The IT administrator suspects that the issue is related to the transport protocol or application-level buffering strategy.

Questions

1. Analyze whether the problem is primarily caused by the **transport protocol (TCP/UDP)** or the **application layer**.
2. Justify your answer based on the communication requirements of real-time multimedia applications.
3. Recommend suitable protocol-level or application-level improvements to enhance conferencing quality.

2.A multinational software company hosts its web services across multiple continents. Employees frequently report long delays before the application homepage loads. Network analysis shows that the **average DNS lookup time is approximately 800 ms**, significantly affecting user experience. The organization plans to redesign its DNS infrastructure while ensuring continuous service availability during server failures.

Questions

1. Analyze the reasons behind the high DNS resolution time.

2. Propose a suitable DNS architecture using techniques such as **Anycast DNS**, **DNS pre-fetching**, and **TTL optimization** to reduce the lookup time below **50 ms**.
3. Evaluate the advantages and limitations of your proposed architecture in terms of performance, scalability, and availability.

3.A cloud service provider delivers applications to millions of users worldwide. During promotional events, users experience intermittent delays in accessing cloud-hosted services. Investigation reveals that DNS queries are taking nearly **800 ms** because all requests are directed to a single primary DNS server. The company wants a highly available DNS solution capable of handling global traffic efficiently.

Questions

4. Examine the impact of centralized DNS architecture on application performance.
5. Design a distributed DNS solution that minimizes lookup latency while maintaining high availability.
6. Analyze how **TTL values**, **Anycast routing**, and **DNS caching** influence system performance and fault tolerance.

4.A financial institution operates an electronic stock trading platform where thousands of buy and sell orders are submitted every second. During market opening hours, the **average order acknowledgment latency increases from 1 ms to 40 ms**, resulting in delayed trade confirmations and customer dissatisfaction. Network engineers observe increased TCP retransmissions, longer connection establishment times, and excessive buffering.

Questions

1. Analyze three possible TCP-related factors responsible for the latency increase, considering **flow control**, **Nagle's algorithm**, and **SYN queue limitations**.
2. Explain how each factor affects transaction processing in high-frequency systems
3. Recommend appropriate TCP configuration changes to reduce latency and improve transaction reliability.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	**25**	**Thorough understanding; clearly identifies all key issues**	**Good understanding with most issues identified**	**Basic understanding; some issues missed**	**Poor understanding; problem not identified correctly**
Application of Concepts	**25**	**Effectively applies relevant concepts to analyze the case deeply**	**Applies appropriate concepts with minor gaps**	**Limited application of concepts**	**Fails to apply relevant concepts**
Analysis	**20**	**In-depth analysis with strong reasoning and insights**	**Good analysis with logical reasoning**	**Basic analysis with limited depth**	**Weak or no analysis; lacks reasoning**
Solution Design	**20**	**Innovative, feasible solutions with strong justification**	**Logical solutions with adequate justification**	**Basic solutions with weak justification**	**Incorrect or no solution provided**
Clarity	**10**	**Clear, wellstructured, and easy to understand**	**Organized and understandable presentation**	**Limited clarity and structure**	**Poor clarity; difficult to understand**

Case Study – Cloud Networking and Transport Protocol Analysis

Question 1: Real-Time Video Conferencing Performance Issues

1. Analyze whether the problem is caused by the transport protocol or the application layer

The issue is **primarily related to the network and transport behavior of UDP-based communication**, but the **application layer buffering strategy also contributes significantly**.

- **8% packet loss** is extremely high for real-time voice/video traffic.
- **180 ms jitter** indicates large variation in packet arrival time.
- **Frozen video frames** and **voice distortion** are typical symptoms of **UDP packet loss and jitter**.
- **Delayed communication** may occur when the application increases its **jitter buffer size** to compensate for unstable packet delivery.

Most enterprise video conferencing platforms use:

- **UDP + RTP (Real-time Transport Protocol)** for audio/video streams.
- **TCP or HTTPS** only for signaling, authentication, and control messages.

Therefore, the **primary root cause is unstable UDP transport conditions**, while the **application layer buffering strategy is a secondary contributing factor**.

2. Justification based on real-time multimedia requirements

Real-time multimedia applications require:

Requirement	Importance
Low latency	Essential
Low jitter	Essential
Continuous playback	More important than perfect reliability
Minimal retransmissions	Essential

TCP is generally unsuitable for live audio/video because:

- Lost packets are retransmitted.
- Retransmissions introduce additional delay.
- Head-of-line blocking causes later packets to wait for missing packets.

UDP is preferred because:

- No retransmission delay.
- Packets are delivered immediately if available.
- Small losses are tolerated using codecs and error concealment.

Given the observed metrics:

- **Packet loss > 1%** already affects voice quality.
- **180 ms jitter** exceeds the acceptable range (usually < 30 ms).
- The conferencing application likely enlarges its **adaptive jitter buffer**, increasing end-to-end delay.

Hence, the behavior is consistent with **UDP-based real-time transport under poor network conditions**, not with TCP retransmission delays.

3. Recommended improvements

Protocol-level improvements

Enable QoS (Quality of Service)

- Prioritize RTP/UDP packets using **DSCP EF (Expedited Forwarding)**.
- Separate conferencing traffic from bulk data transfers.

Use RTP with RTCP

- RTP carries media streams.
- RTCP provides feedback about packet loss, jitter, and delay.

Implement Forward Error Correction (FEC)

- Adds redundant packets.
- Allows recovery from small packet losses without retransmission.

Adopt adaptive bitrate codecs

- H.264/H.265 with dynamic bitrate adjustment.
- Opus codec for audio.

Application-level improvements

Adaptive jitter buffering

- Maintain a buffer between **20–50 ms** under normal conditions.
- Avoid excessively large buffers that increase conversational delay.

Packet loss concealment (PLC)

- Interpolate missing audio packets.
- Prevent noticeable voice distortion.

Dynamic resolution and frame-rate scaling

- Reduce video resolution during congestion.
- Lower frame rate before allowing complete frame freezing.

Bandwidth estimation

- Continuously monitor available bandwidth.
- Adjust encoding rate proactively.

Expected outcome

Improvement	Expected Benefit
QoS + RTP prioritization	Reduced packet loss
FEC	Better voice continuity
Adaptive jitter buffer	Reduced delay
Adaptive bitrate	Fewer frozen frames
RTCP feedback	Faster congestion response

With these changes, packet loss can typically be reduced below **1%**, jitter below **30 ms**, and conferencing quality becomes suitable for enterprise use.

Question 2: High DNS Resolution Time in a Global Web Service

1. Reasons for the 800 ms DNS lookup time

An average **800 ms DNS resolution time** is unusually high and indicates architectural problems.

Possible causes

Single centralized DNS server

- All continents must contact one distant server.
- Long round-trip propagation delays occur.

Recursive resolver chaining

- Multiple recursive queries increase latency.

Low TTL values

- Resolvers cannot cache responses effectively.
- Frequent authoritative lookups occur.

Lack of geographically distributed authoritative servers

- Users in Asia may query servers located in Europe or North America.

DNSSEC overhead (if poorly optimized)

- Additional signature validation increases response time.

Network congestion or overloaded DNS servers

- High query volume increases processing delay.

2. Proposed DNS architecture (< 50 ms target)

Distributed Anycast DNS Architecture

```
Users
  |
Local ISP Resolver
  |
Anycast Authoritative DNS
 /    |     \
Asia   Europe  America
```

Key techniques

a) Anycast DNS

- Multiple DNS servers share the **same IP address**.
- BGP routes users to the **nearest healthy server**.

Expected latency

- Regional users: **10–30 ms**
- Cross-region fallback: **30–50 ms**

b) DNS Pre-fetching

Web applications can trigger DNS resolution **before the user clicks a link**.

Example:

<link rel="dns-prefetch" href="//api.company.com">

This reduces perceived page-load delay.

c) TTL Optimization

Record Type	Recommended TTL
A / AAAA	300–600 s
CDN records	60–120 s
MX records	3600 s
NS records	86400 s

This balances **caching efficiency** and **failover responsiveness**.

Additional enhancements

- Deploy **regional DNS clusters** in Asia, Europe, and America.
- Use **health checks** with automatic route withdrawal.
- Enable **EDNS Client Subnet (ECS)** for location-aware responses.
- Use **DNS over UDP** with TCP fallback only when necessary.

3. Advantages and limitations

Advantages

Aspect	Benefit
Performance	Nearest DNS server reduces latency
Scalability	Traffic distributed across regions
Availability	Server failures do not interrupt service
Load balancing	Queries spread automatically
User experience	Faster page loads

Limitations

Aspect	Limitation
Anycast management	More complex BGP configuration
TTL tuning	Very low TTL increases query volume
Cache inconsistency	Different resolvers may hold stale records
Operational cost	Multiple global DNS PoPs required

Evaluation

Metric	Centralized DNS	Proposed Architecture
Average latency	~800 ms	20–40 ms
Scalability	Low	High
Fault tolerance	Poor	Excellent
Geographic optimization	None	Strong
Operational complexity	Low	Moderate

The proposed architecture is capable of meeting the **< 50 ms DNS lookup requirement** while maintaining continuous availability during server failures.

Question 3: Distributed DNS for a Global Cloud Provider

4. Impact of centralized DNS architecture

A centralized DNS design negatively affects cloud-hosted applications in several ways.

Performance impact

High lookup latency

- Every user must reach a single distant DNS server.

Increased page-load time

- DNS resolution occurs before TCP/TLS connection establishment.

Server bottlenecks

- Promotional events generate millions of simultaneous queries.
- CPU, memory, and network resources become saturated.

Single point of failure

- If the primary DNS server fails, all cloud services become unreachable.

Poor geographic scalability

- Users far from the DNS server experience the highest delays.

5. Distributed DNS solution

Components

Authoritative DNS clusters

- Multiple servers per region.
- Active-active operation.

Anycast routing

- Same IP advertised globally.
- BGP directs users to the closest available cluster.

Recursive caching layer

- ISP and enterprise resolvers cache responses.
- Reduces authoritative query load.

Health monitoring

- Failed nodes automatically stop advertising routes.
- Traffic shifts to healthy regions within seconds.

Result

Region	Typical Lookup Time
Asia users	15–25 ms
Europe users	10–20 ms
America users	10–20 ms

6. Influence of TTL, Anycast, and DNS caching

TTL values

Low TTL (30–60 s)

Advantages

- Fast failover.
- Rapid traffic redirection.

Disadvantages

- More DNS queries.

- Higher authoritative server load.

Moderate TTL (300–600 s)
Best balance for cloud services:
- Good caching efficiency.
- Acceptable failover time.

Anycast routing
Performance benefits
- Routes queries to the nearest DNS node.
- Minimizes round-trip delay.
- Automatically distributes global traffic.

Fault tolerance
- If a regional server fails, BGP converges to the next nearest node.
- No client-side reconfiguration is required.

DNS caching
Caching occurs at:
- Browser cache
- Operating system cache
- ISP recursive resolver
- Enterprise resolver

Effects

Effect	Impact
Reduced latency	Immediate responses from cache
Lower bandwidth usage	Fewer external DNS queries
Reduced server load	Better scalability
Improved resilience	Temporary operation continues even if authoritative servers are unreachable

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Overall analysis

Mechanism	Performance	Scalability	Fault Tolerance
TTL optimization	Medium	Medium	High
Anycast routing	High	High	High
DNS caching	Very High	Very High	Medium

Combining **Anycast + caching + optimized TTLs** provides the best overall balance for large-scale cloud platforms.

Question 4: TCP Latency in an Electronic Stock Trading Platform
1. Three TCP-related factors causing latency increase
Factor 1: TCP Flow Control (Receive Window Limitation)

TCP uses a **receive window (rwnd)** to prevent the sender from overwhelming the receiver.

During market opening:

- Thousands of acknowledgments arrive simultaneously.
- Application threads may process data slowly.
- Receive buffers fill up.
- The advertised window shrinks.

Effect

- Senders pause transmission.
- Order acknowledgments wait in kernel buffers.
- Latency increases from **1 ms to tens of milliseconds**.

Factor 2: Nagle's Algorithm

Nagle's algorithm combines small packets to reduce network overhead.

Stock trading orders are typically **very small messages** (tens or hundreds of bytes).

With Nagle enabled:

1. A small order is sent.
2. TCP waits for its ACK before sending the next small packet.
3. Multiple orders may be delayed unnecessarily.

Effect

- Microsecond-level delays accumulate.
- Interactive order/acknowledgment exchanges become slower.
- High-frequency trading systems are particularly sensitive to this behavior.

Factor 3: SYN Queue Limitations

Each new TCP connection requires a **three-way handshake**.

During market opening:

- Many clients establish connections simultaneously.
- The server's **SYN backlog queue** may become full.

Consequences:

- SYN packets are dropped.
- Clients retransmit SYN requests.
- Connection establishment time increases significantly.

Effect

- New trading sessions experience delays.
- Order submission is postponed until the connection is established.

2. Effect on high-frequency transaction processing

TCP Factor	**Effect on Trading System**
Flow control	Delays acknowledgment delivery
Nagle's algorithm	Introduces artificial waiting for small packets
SYN queue limitation	Slows connection establishment

In high-frequency trading:

- Orders must be acknowledged within **microseconds or low milliseconds**.
- Even a **40 ms delay** can affect trade execution priority and customer confidence.

3. Recommended TCP configuration changes

a) Disable Nagle's Algorithm

setsockopt(socket, IPPROTO_TCP, TCP_NODELAY, ...);

Benefit

- Small order packets are transmitted immediately.
- Reduces interactive latency dramatically.

b) Increase Socket Buffer Sizes

Linux example:

sysctl -w net.core.rmem_max=134217728
sysctl -w net.core.wmem_max=134217728
sysctl -w net.ipv4.tcp_rmem="4096 87380 134217728"
sysctl -w net.ipv4.tcp_wmem="4096 65536 134217728"

Benefit

- Prevents receive-window exhaustion.
- Reduces flow-control pauses.

c) Expand SYN Backlog Capacity

sysctl -w net.ipv4.tcp_max_syn_backlog=65535
sysctl -w net.core.somaxconn=65535

Benefit

- Handles large bursts of simultaneous connection requests.
- Reduces SYN retransmissions.

d) Enable SYN Cookies

sysctl -w net.ipv4.tcp_syncookies=1

Provides protection when backlog queues overflow.

e) Use Persistent Connections

- Maintain long-lived TCP sessions.
- Avoid repeated three-way handshakes for every order.

f) Reduce Bufferbloat

Use modern queue disciplines such as **fq** or **fq_codel**:

tc qdisc replace dev eth0 root fq

This minimizes excessive buffering delay.

Expected improvement

Optimization	Expected Latency Reduction
TCP_NODELAY	5–15 ms
Larger buffers	5–10 ms
Increased SYN backlog	10–20 ms
Persistent connections	1–5 ms
Queue management	2–8 ms

Combined, these changes can typically reduce acknowledgment latency from **40 ms back to approximately 1–3 ms**, which is suitable for electronic trading environments.

Conclusion

The four case studies demonstrate that **transport protocols, DNS architecture, and TCP tuning are critical determinants of application performance in modern distributed systems**.

- **Real-time multimedia applications** should prioritize **UDP/RTP with adaptive buffering and QoS mechanisms** rather than relying on TCP reliability.
- **Global web and cloud platforms** require **distributed Anycast DNS architectures, intelligent caching, and optimized TTL values** to achieve sub-50 ms resolution times and high availability.
- **Centralized DNS infrastructures** create significant latency, scalability, and fault-tolerance problems that can be eliminated through geographically distributed authoritative DNS clusters.
- **High-frequency financial systems** demand **low-latency TCP configurations**, including **TCP_NODELAY, larger socket buffers, expanded SYN backlogs, persistent connections, and modern queue management techniques**.

By applying these networking principles, organizations can significantly improve **performance, scalability, reliability, and user experience** across enterprise conferencing systems, global cloud services, and mission-critical financial transaction platforms.